

Class Test 4 (21 September 2026)

- A set S of real numbers is *bounded* if and only if there exists a real number K such that $|x| \leq K$ for any $x \in S$.
 - *Continuum Property*: Every non-empty set of real numbers which is bounded above has a smallest upper bound. Every non-empty set of real numbers which is bounded below has a largest lower bound.
 - The smallest upper bound is called the *supremum* of the set.
 - The largest lower bound is called the *infimum* of the set.
 - We say that a sequence $\{x_n\}$ *diverges* to $+\infty$ and write $\{x_n\} \rightarrow +\infty$ as $n \rightarrow \infty$ if, for any $H > 0$, we can find an N such that, for any $n > N$, $x_n > H$.
- (a) Let S be a non-empty set of real numbers which is bounded above. Show that a sequence of points of S can be found which converges to its supremum.
- (b) Let S be a non-empty set of real numbers which is unbounded above. Show that a sequence of points of S can be found which diverges to $+\infty$.
- (c) Let C be a *compact* subset of $\mathbb{R}^n$ and $f : C \rightarrow \mathbb{R}$ be *continuous* on C . Prove that f is bounded on C .
- (d) Let C be a *compact* subset of $\mathbb{R}^n$ and $f : C \rightarrow \mathbb{R}$ be *continuous* on C . Prove that there exist points x_m and x_M in C such that $f(x_m) \leq f(x) \leq f(x_M)$, for all $x \in C$.